

Section 6.2: The Effects of ICT on Society

ICT has had many impacts upon our lives both for the better and for the worse.

This section looks at some of these impacts that ICT has had upon society and includes **impacts on employment, online shopping, how computers are used in household appliances** and how computers have given us the ability to **never actually leave our houses**.

Key Concepts of this section:

- # Understand the effects that ICT has had on **employment patterns**.
- # Know the **effects** that **microprocessor-controlled devices** can have upon our lives.
- # Be able to discuss how ICT can affect our lives both **positively** and **negatively**.

Impacts on Patterns of Employment

Why do businesses use ICT?

- # Computer usage in businesses and companies has increased massively within the last 30 years. The benefits that computers bring to business is just too hard for the owners to ignore. For example:

1. **Speed** → Data can be **stored, processed** and **searched** extremely fast
2. **Volume** → Vast quantities of data can be stored in small amounts of space
3. **Less Paper Work** → Computers **remove the need** for huge quantities of **paper-work**
4. **Less Workers** → Computers can do the **work of many people** which **saves expenses in wages**



Key Words:

Automation, Negative Effects, Positive Effects, Viewpoint



Before computers there had to be lots of workers to run a business

5. Connectivity ➔ **The Internet** means that companies can **share data** or even **hold meetings** in different countries **without actually leaving the office**



One person and a computer can do the job of many manual workers

However, the heavy use of ICT does have certain effects upon employment. These effects can be positive or negative depending upon your viewpoint.

What effects has ICT had upon employment?

Effects of ICT on employment patterns include:

- **Fewer workers needed to achieve the same amount of work**
- **Automation in factories (robots doing the work of people)**
- **Increased 'teleworking' (working from home)**
- **Increased need for highly trained ICT specialists**
- **Decreased need for 'labor intensive' workers (like filing, printers etc)**
- **Need for workers to be trained more frequently (to keep up with the changes in ICT)**
- **Fewer 'face-to-face' meetings as video conferences can be held over the Internet (reduced travel time and cost)**

NOTE!

These effects can be **positive** or **negative** depending on your viewpoint. For example:

A factory boss would like automation because robots can work harder and are cheaper than people.



However the people who lose their jobs to the robots will not be too happy.



Areas where jobs have been created

ICT has brought new jobs to the world These include:

1. Network Managers / Technicians

These jobs have been created due to the need for big **computer networks within companies**. Network managers make sure that computers on the network **run smoothly** and are **secure**.

2. Web Design

The Internet has seen a huge demand for both **commercial (business) and personal web sites**. Web designers create these sites and keep them up to date.

3. Systems Analyst

These workers are responsible for designing **new ICT systems**.



Web Design jobs have been created due to ICT

4. Programmers

Programmers create all the **software that can be used on a computer**. This is a huge industry and produces things like **operating systems, applications** (word, excel etc) and **computer games**.

5. Computer Engineers

Engineers are needed to **build the high-tech computers** that we use today.



Gaming industries have exploded in recent year

Areas where jobs have been lost

Unfortunately ICT has also caused the loss of jobs in certain areas. These areas include:

1. Manual Repetitive Jobs

Factory manufacturing, spraying, welding, packing and assembling have seen **robots replacing humans**. Robots are replacing people because they can do the same job much **faster**, they can **work 24/7** and they **do not require wages**.

2. Shop Work

Online shopping (**like Amazon**) has caused a **reduction in the number of shops** found on our high streets. More and more people are shopping online because it is generally **cheaper** and they can get **goods delivered**.



Robots do the work of many people

3. Bank Cashiers

Use of **ATM's** have reduced the need for bank cashiers.

4. Office Work

Traditional office work such as **typists** and **filers** have been replaced by computers. One person who knows how to use **Word Processors** and **Databases** can replace many typists and filers.

5. Telephone Exchange

Telephone operators have been replaced by computer systems that make **telephone connections automatically**.

6. Library Work

With the launch of e-books (electronic books) the future of traditional libraries could be at risk. People could easily download an electronic book instead of having to visit a library.



ATM's reduce the need for bank cashiers

Microprocessor Controlled Devices

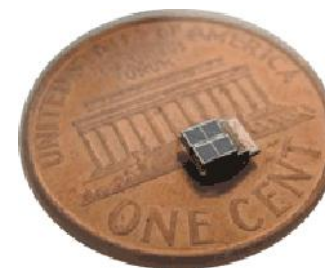
Key Words:

What is a microprocessor?

- # A microprocessor is really just a **tiny cpu built onto a single chip**.
- # Microprocessors are put into **electronic devices** to **control how they function**.
- # Microprocessor controlled devices affect our lives in lots of different ways including in **our homes**, our **leisure time**, how we **interact socially**. They even have an affect on our needs to actually **leave our houses**.



Virus, Replicate, Infect,
Crash, Pop-Up,
Prevention



A tiny microprocessor

Effects of microprocessors in the home

- # Any electronic device in your house that needs to be controlled in some way is likely to contain a microprocessor. Examples of these are listed below:

1 Computer System

Your desktop PC or laptop will contain a microprocessor called a **CPU** (powerful machines will contain more than one).

The CPU **controls all the processing** that go's on within the computer.



2 Washing Machines

Washing machines contain microprocessors to control things like:

- **Water temperature**
- **Valves to let water in**
- **Pumps to let water out**



3 Alarm Systems

The microprocessor inside a burglar alarm is used to **detect intruders** and they also **set the alarm off**.

Some even contact the police for you automatically.



4 Heating Systems

Heating systems use microprocessors to control things like:

- **Time when the heating is switched on / off**
- **Keeping a constant temperature**
- **Some can even make the temperature different from room to room**



5 Intelligent Ovens / Microwaves / Fridges

The microprocessor inside intelligent ovens can **read the barcodes on packaging** and then automatically **set the temperature** and **cooking time** to the perfect levels.

Intelligent microwaves work in much the same way.

Intelligent fridges can read the barcodes on food and determine the '**use by**' dates. Food that is due to go bad is then **automatically moved to the front of the fridge** so that you use it first.



Effects of microprocessors on leisure time

- # Microprocessor-controlled devices allow us to do our **daily tasks quicker**. This gives us **more time for leisure** and other things that we might enjoy.

For example we could set the washing machine to wash and then dry our clothes while we read a book or watch a movie.

- # Other ways in which microprocessors can affect how we spend our leisure time are listed below:

1 Working on the move

Laptops, PDA's, Mobile Phones, iPads etc, allow us to **work on the move** which means we get to **enjoy more free time**.

For example, I could complete my work for the evening using a laptop on the train. This means I don't have to do it when I get home.



2 More time spent gaming

There has been a huge **increase in the demand for gaming** over the last 20 years. More and more people spend a large amount of free time playing the latest computer games.



3 Increased leisure time

Microprocessors in **household appliances** can make sure that the **task is completed without you actually being there**.

Once you have started the appliance you can go off and do something else and return to it later.



4 Lack of Fitness

Because microprocessor-controlled devices **do most of the work for us** it means that we are **not doing as much hard manual work** as we used to.

For example, before we had washing machines, doing the laundry was actually quite a demanding task. You had to wash the clothes by hand and then hang them on a washing line. Now all we need to do is put them in the machine and press 'go'.



5 Increased Fitness

Because microprocessor-controlled devices can **reduce the amount of time** it takes us to **complete our every-day tasks** we are left with **more free time** to do the things we want to do.

Some people choose to spend some of this extra time in the **gym getting fit**.



Effects of microprocessors on social interaction

Microprocessor-controlled devices like **computers and mobile phones** have made it easier for use to **contact and stay in touch** with friends and family.

For example, if we want to chat to a friend we can either **call/text** them on a mobile, **skype them, chat on msn** or even **email**.

Examples of microprocessor-controlled devices that we use to interact with:

1 Mobile Phones

Mobile phones allow us many ways in which to interact or chat to friends and family. These ways include:

- **Calling**
- **Text / SMS**
- **Email facilities**
- **Accessing social networking site and interacting there**

- **Accessing chat rooms and chatting there**
- **Instant messaging (Blackberry Messenger - BBM).**



2 Social Networking Sites

Social networking sites are growing more and more popular as a way to **keep in contact with friends**. Facebook alone has almost 1 billion active users.

These sites have many ways in which you can interact including:

- **Private messaging**
- **Status updates**
- **Emails.**



3 VoIP service such as Skype

Skype is known as a '**VoIP**' (**voice over Internet protocol**) service which allows you to make **very cheap phone calls over the Internet**.

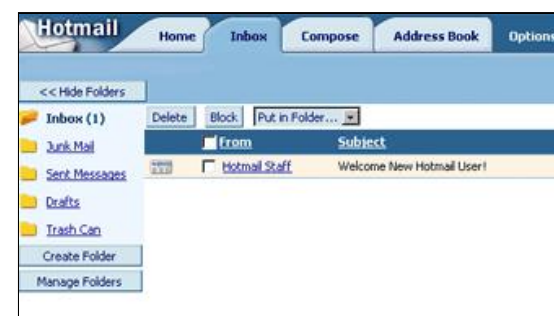
Skype is especially useful if you want to talk to someone in another country as the prices are very cheap.



4 Emails

Good old emails are still **sent in their billions**. They are quick and easy and allow us to **send and receive messages**, no matter where we are in the world, at the click of a button.

Emailing can now be **carried out on the move** thanks to **portable devices** (like mobile phones, iPads and laptops) and **Wi-Fi**.





Positive effects of microprocessors on social interaction

1. Easy to make **new friends** in chat-rooms.
2. Easy to find people with the **same interests or hobbies** as you in specialist chat-rooms.
3. **Cheap Internet phone calls** can be made using VoIP.
4. Easy to **spread the news of social events** using social networking.
5. **Old or disabled people** can **still interact** without having to leave their homes.



Negative effects of microprocessors on social interaction

1. You don't actually meet people face to face as much. This is known as '**social-isolation**'.
2. Interacting through a computer **does not require much physical activity** and can lead to health problems like **obesity**.

Effects of microprocessors on the need to leave the home

- # ICT has progressed to a point where it would be possible to never have to leave your home.
- # Many **everyday tasks** can be carried out at home using microprocessor-controlled devices.

Examples of how microprocessors can affect the need to leave the house:

1 Online Shopping (E-Commerce)

Goods (of any kind) can be **ordered over the Internet** and **delivered to your door**.

Any device that allows you to connect to the Internet can be used to shop online.

Retailers that offer customers the ability to buy items online include:

- **Supermarkets**
- **Entertainment businesses (DVD, Music, Games)**
- **Clothing.**



2 Teleworking (working from home)

Many people **work from home** and use ICT to **perform tasks and contact clients and colleagues**.

Teleworking requires the following equipment:

- **Computer or Laptop**
- **Internet connection**
- **Email facility**
- **Videoconferencing software (meetings help over the Internet)**



3 Entertainment

We can entertain ourselves very easily using microprocessor devices and the Internet. For example:

- **YouTube (videos, music etc)**
- **Gaming**
- **Social networking to chat to friends**
- **Downloading mp3's to the computer**
- **Downloading and watching DVD's**
- **Chat-rooms.**



4 Online Banking

Banking can now be **carried out online**. We can perform all sorts of tasks without actually visiting the bank. These tasks include:

- **Transferring money**
- **Viewing statements to track your spending**
- **Paying bills**
- **Setting up new accounts**
- **Applying for loans.**



5 Research

Research can be **carried out using a computer with an Internet connection**. For example we can find the following information online:

- **Bus timetables**
- **Shop opening times**
- **Movie showing times**
- **Information for a school assignment.**



Positive effects of not having to leave the house

1. **Elderly or disabled** people can still carry out day-to-day tasks such as **shopping** and **paying bills**.
2. You can find entertainment even if you cannot go outside due to **bad weather**.
3. You can carry out **day-to-day tasks** even **if you are sick**.



Negative effects of not having to leave the house

1. You don't actually meet people face to face as much. This is known as '**social-isolation**'.
2. Not leaving the house is bad for you - **lack of exercise** etc.

Activity!

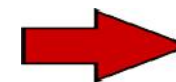
Effects of ICT on Society - Questions

Click the above task and follow the instructions given on the sheet. Make sure that you complete all questions in full.



[Previous - Negative Aspects of ICT](#)

[Next - Capabilities and Limitations of ICT](#)



Links to Theory Units:

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